

CYBER THREATS

And Their Various Types

With so many government, commercial, defence and private systems connected to the ubiquitous Internet, cyber threats are becoming a major concern. It is important for everyone to know about various types of cyber threats in order to safeguard the confidentiality, integrity, authenticity, reliability and availability of data transmitted over the interconnected networks. Here is an overview

DR RAJIV KUMAR SINGH

The use of information technology (IT) has changed our lives dramatically. Nearly all our basic societal activities and daily routines like financial transactions, ticketing, hotel booking, communication with others and working of any organisation depend on the functionality of the digitally operated and interconnected computer world.

This digital world, with or without the use of interconnected networks, is threatened by cyber criminals who try to steal important data about the user (such as credit card numbers, bank account information, user ID and password). Besides, they can damage information on the user's computer, install unwanted software without the user's consent, allow someone else to control the user's computer remotely, use the user's computer resources (such as hard disk space, fast processor and Internet connection) to attack other computers on the Internet), etc. In some cases, intruders take advantage of a particular vulnerability in one of the programs installed in the computer to gain access.

Given these risks, it is very important for everyone to have knowledge about cyber threats in order to safeguard the confidentiality, integrity, authenticity, reliability and availability of data transmitted over interconnected networks.

Cyber threats

Cyber threats are usually classified as passive and active attacks. Passive attacks try to eavesdrop to learn contents

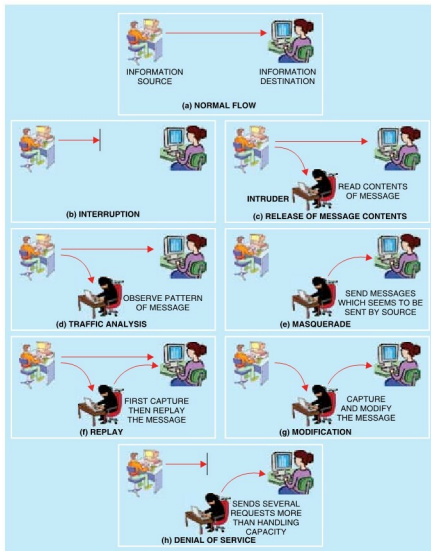


Fig. 1: Communication flow between source and destination and possible types of cyber attacks

or make use of information from the system without affecting the contents and system resources. In this way, passive attacks may be used for release of message contents and traffic analysis (observe pattern of messages, traffic

flows, etc).

Active attacks, on the other hand, attempt to alter system resource, or affect their operation, in addition to modification of content and/or participation in communication in order